

Physical Chemistry (I)

Lecture 21

Chemical Equilibria



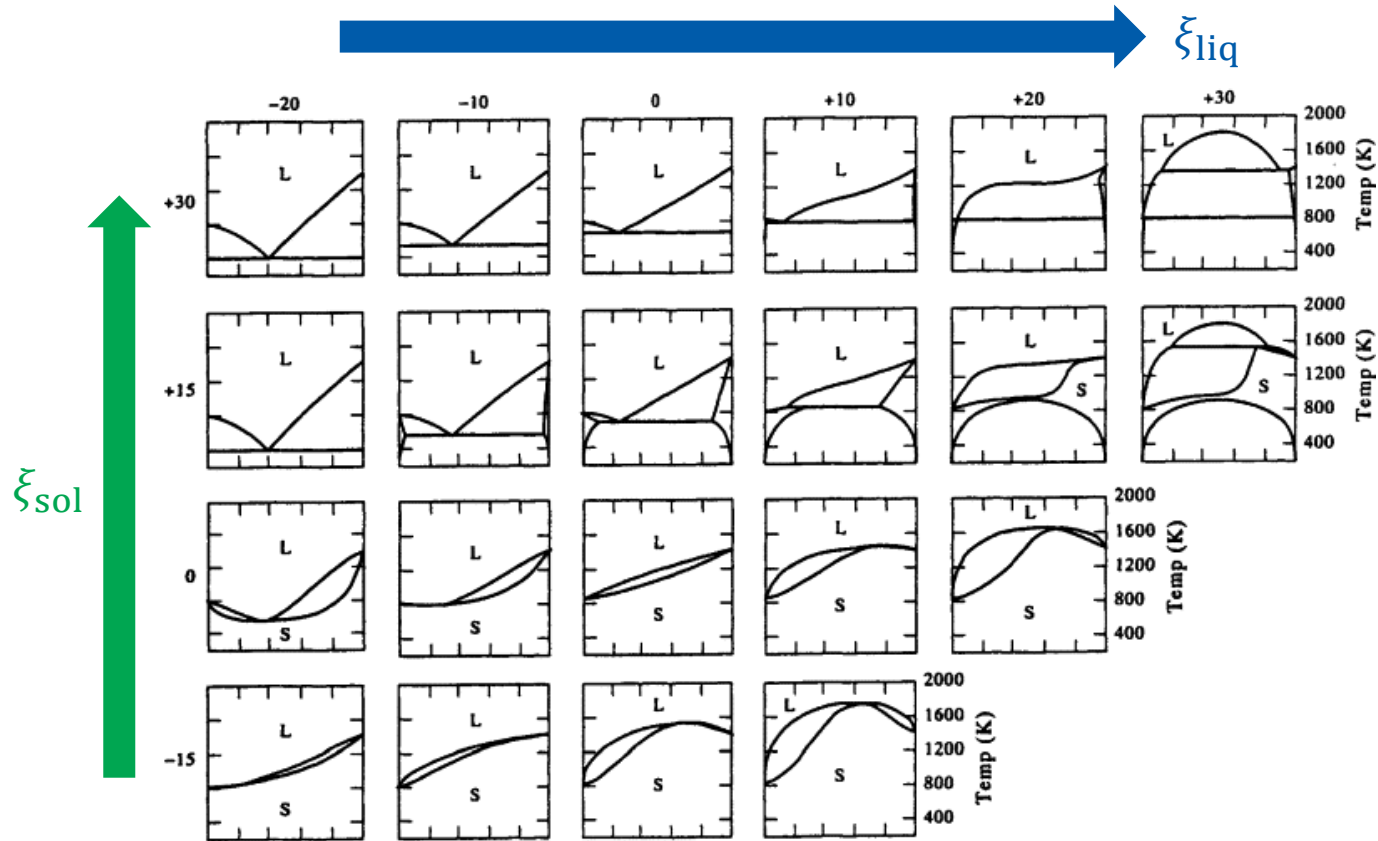
In the Last Class

- Phase diagrams of liquid-solid mixtures
 - Simple systems
 - Reacting systems
- Colligative properties
 - Thermodynamic origin
 - Derivations
- Phase diagrams of ternary systems



Last class

Liquid-Solid Phase Diagrams

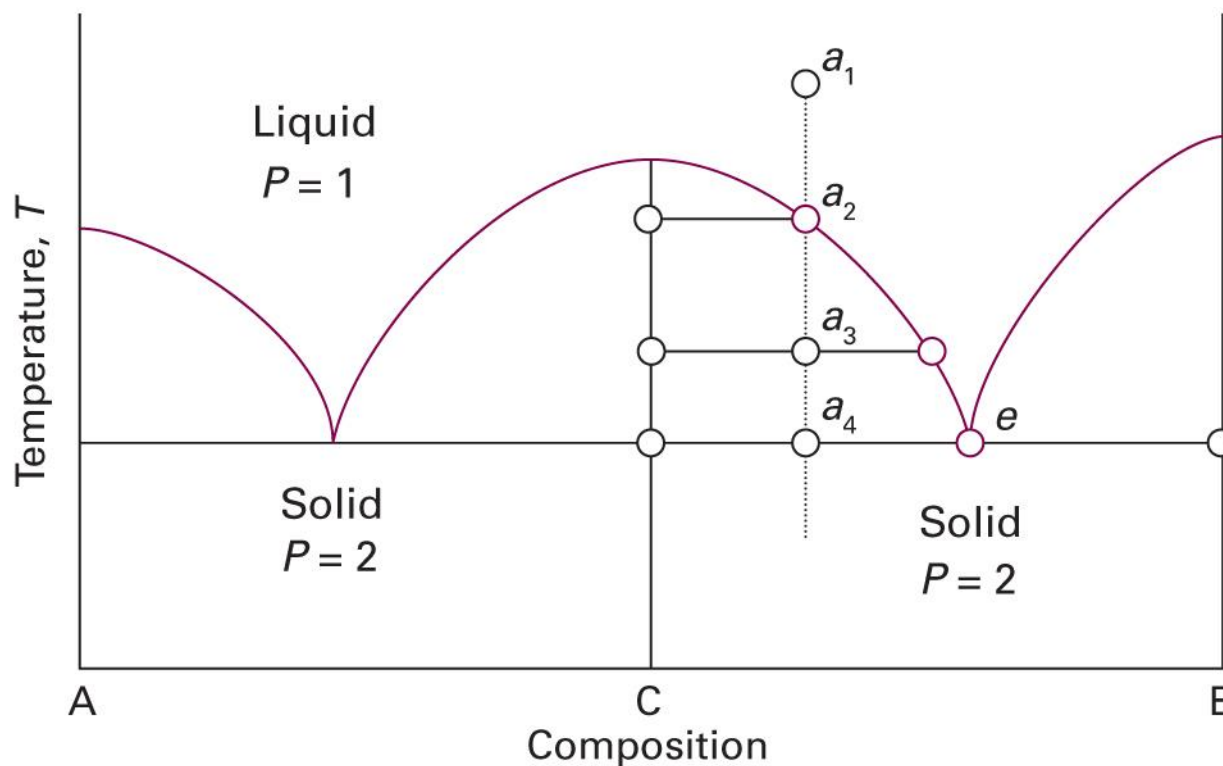


(Image: Gaskell, *Introduction to the Thermodynamics of Materials*, 5th Ed., Fig. 10.23)

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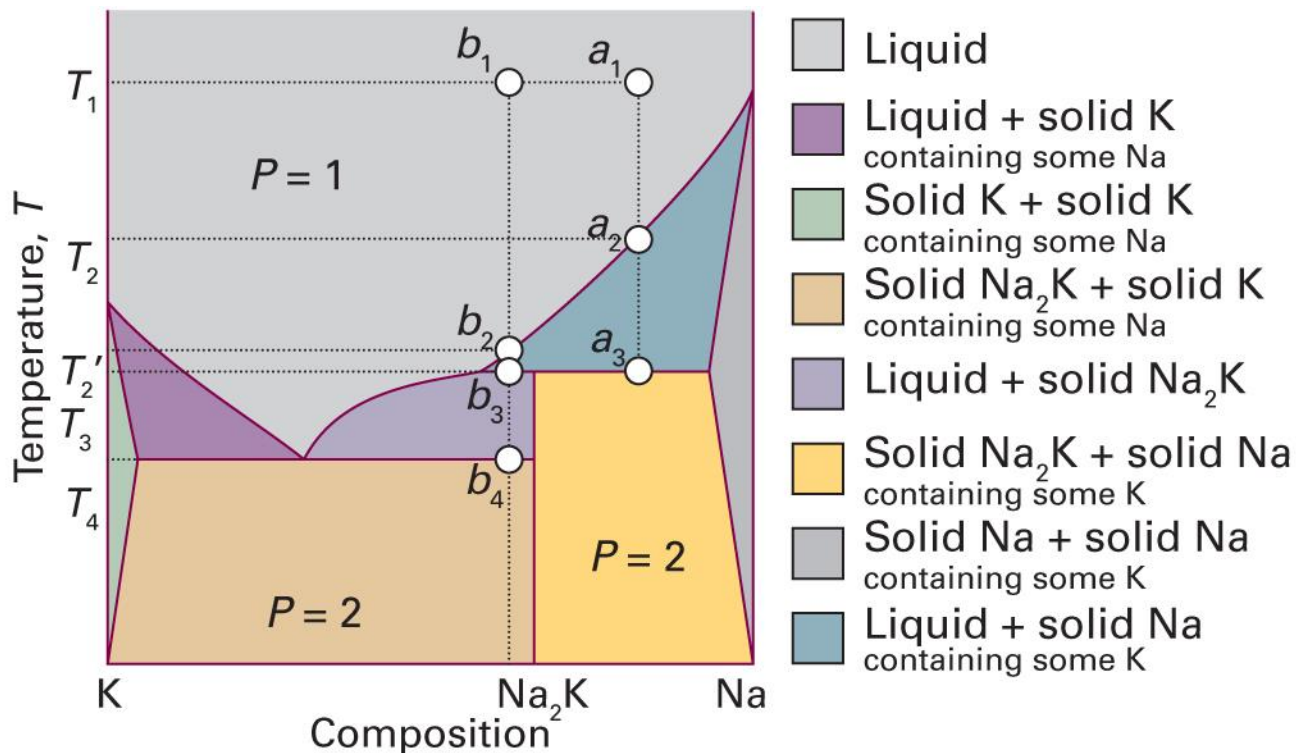
Congruent Melting

(assuming completely immiscible solid)



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Incongruent Melting



(Image: Atkins et al., *Atkins' Physical Chemistry*, 11th ed., Figure 5D.5)

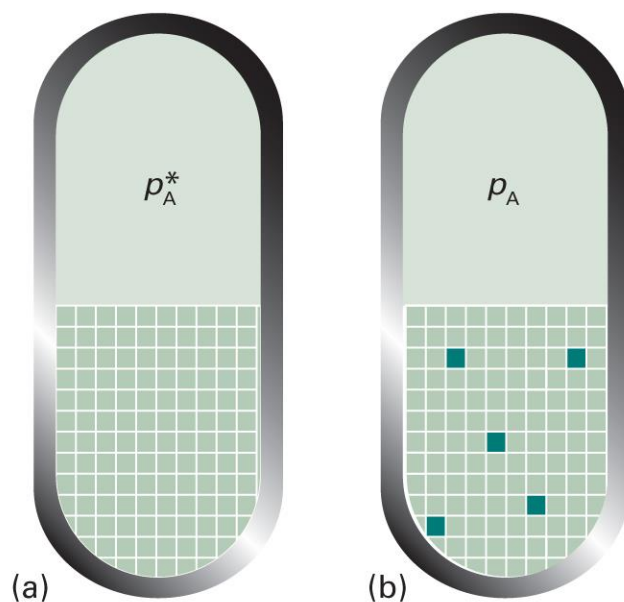
Colligative Properties

- Physical properties that depends on the relative number of solute particles present but not their chemical identity
1. Lowering of vapor pressure
 2. Elevation of boiling point
 3. Depression of freezing point
 4. Osmotic pressure

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Origin of Colligative Properties

- Due to the **reduction of the chemical potential** of the liquid solvent as a result of the presence of solute

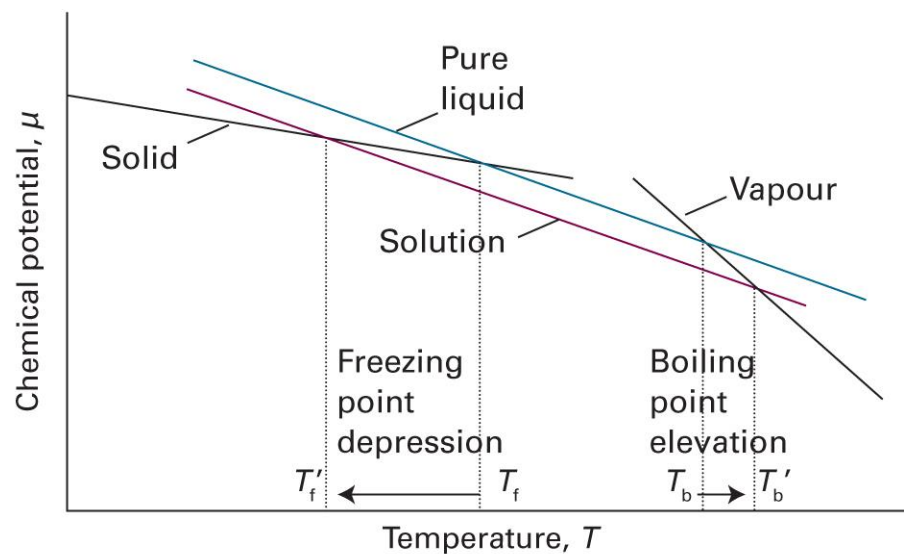


(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5B.7)

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Changes in Phase Transitions

$$\mu_A = \mu_A^* + RT \ln x_A \quad (x_A < 1)$$

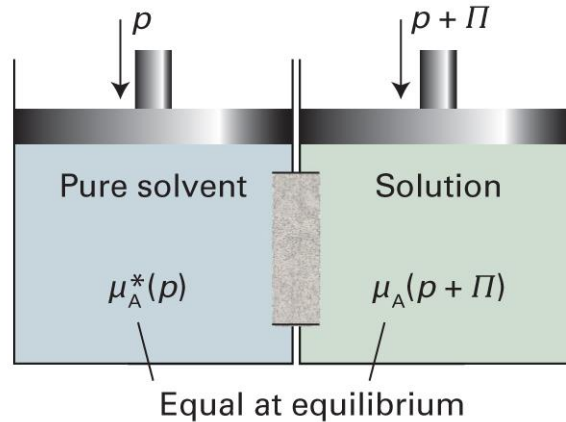


(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5B.6)

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Osmosis

the spontaneous passage of a pure solvent into a solution separated from it by a semipermeable membrane



van't Hoff Factor



- van't Hoff factor i

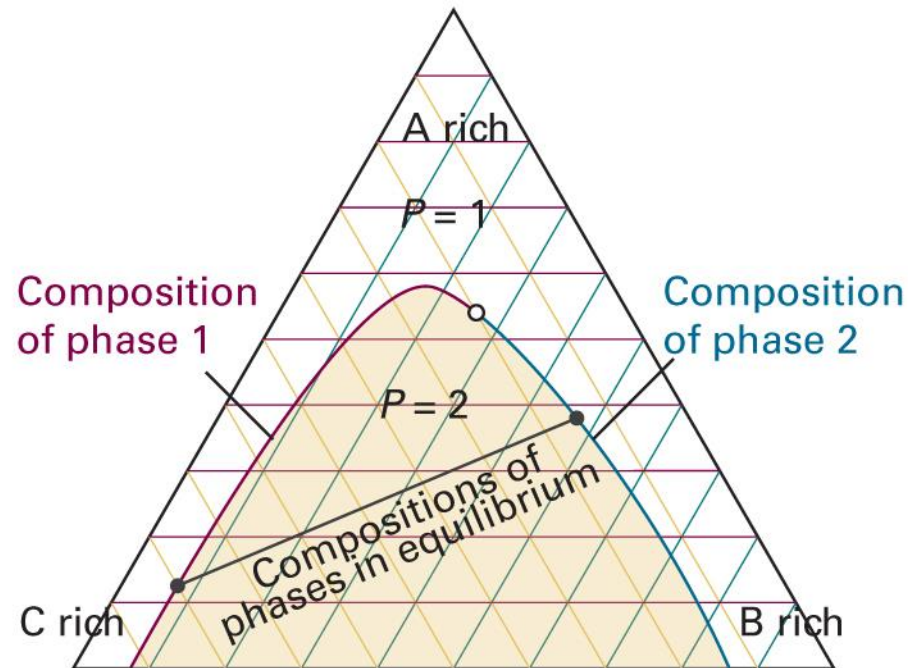
$$\Delta T_b = iK_b b$$

$$\Delta T_f = iK_f b$$

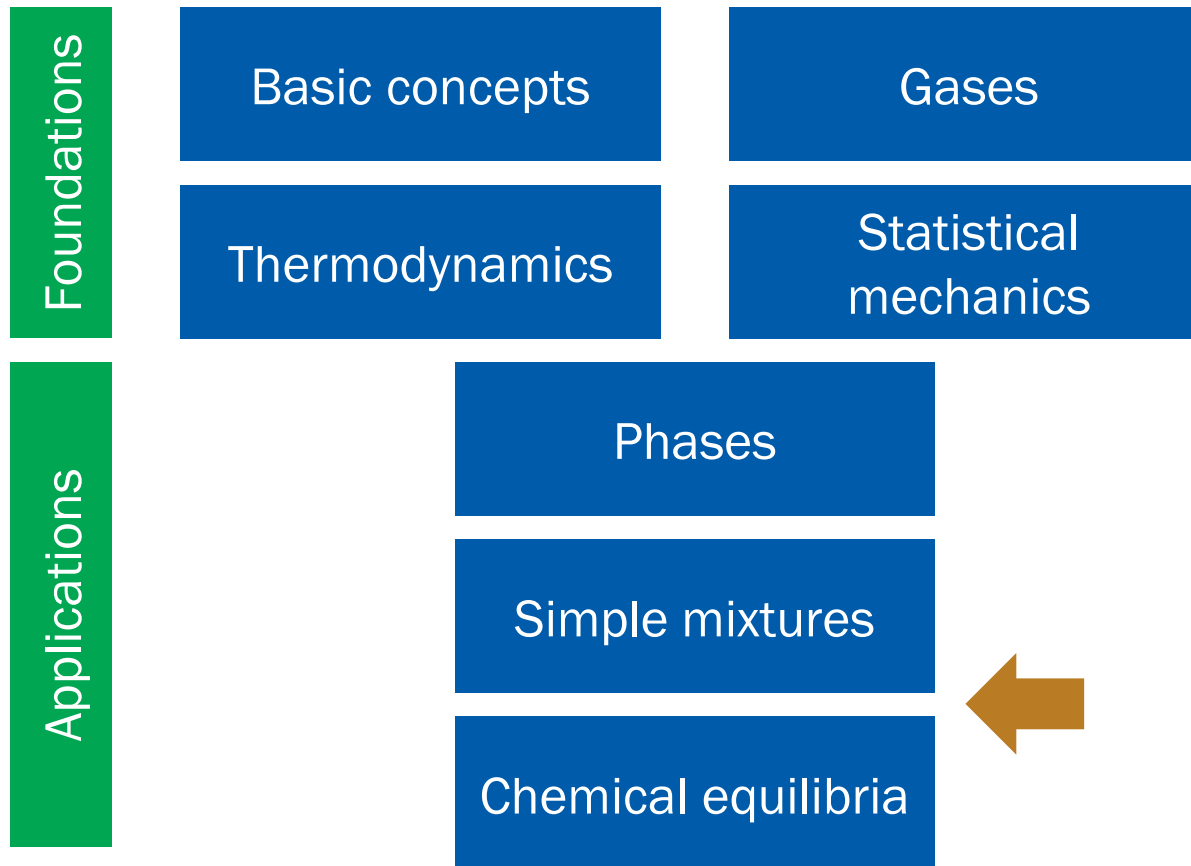
$$\Pi = i[B]RT$$

Last class

Ternary Phase Diagram



Where Are We?



9. Chemical Equilibria

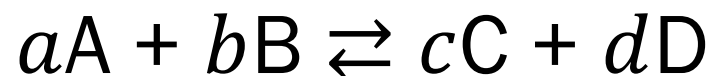


Chapter Overview

- Chemical reactions and equilibria
 - Reaction quotient and equilibrium constant
 - Activity approximations
 - Gaseous reactions
- Le Chatelier's principle



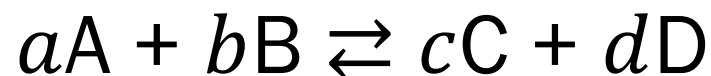
Chemical Reactions



Can we construct
thermodynamics of chemical reactions?

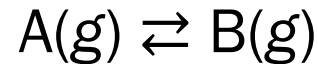
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Extent of Reaction ξ



Species	Initial	→	Change	→	Final
A	n_A		$-a\xi$		$n_A - a\xi$
B	n_B		$-b\xi$		$n_B - b\xi$
C	n_C		$+c\xi$		$n_C + c\xi$
D	n_D		$+d\xi$		$n_D + d\xi$

Simple Reaction



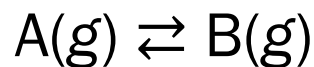
- Extent of reaction ξ : for $d\xi$, $dn_A = -d\xi$ and $dn_B = +d\xi$
- Gibbs energy $G = G(p, T, n_A, n_B)$

For constant p and T ,

$$dG = \mu_A dn_A + \mu_B dn_B = -\mu_A d\xi + \mu_B d\xi = (\mu_B - \mu_A) d\xi$$

$$\left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \mu_B - \mu_A$$

Reaction Gibbs Energy



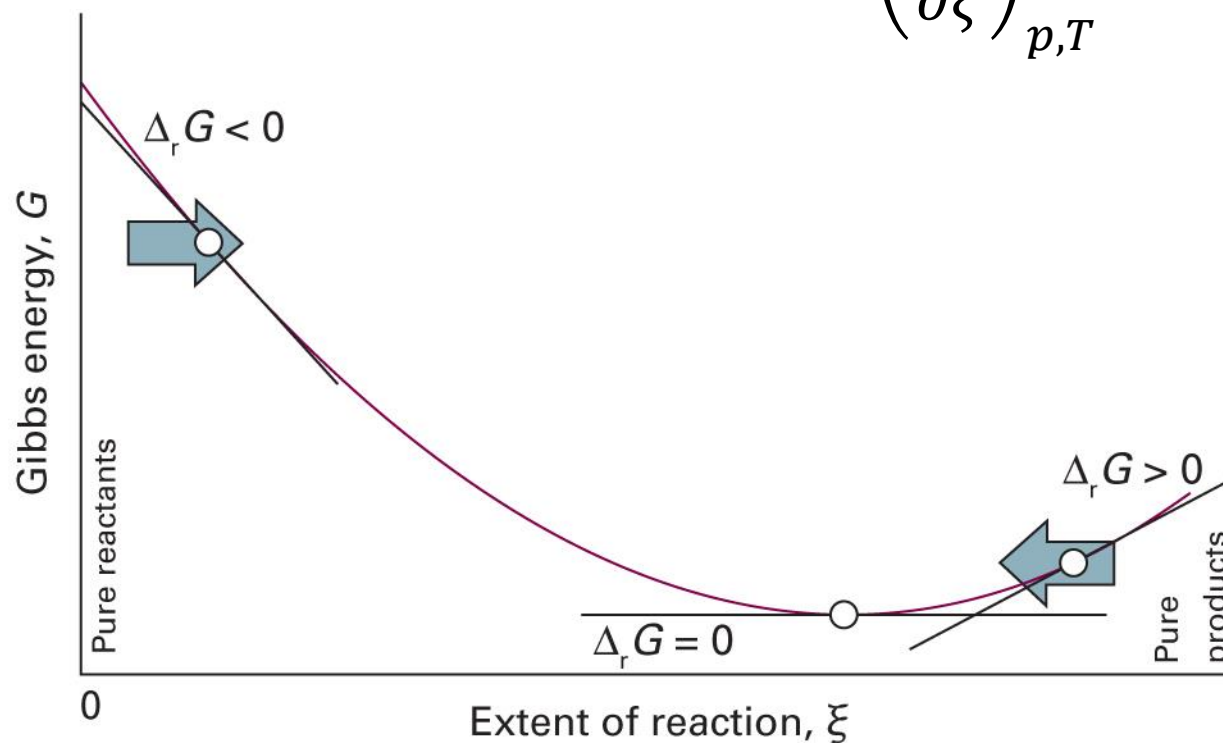
Assuming the absolute scale,

$$\Delta_{\text{r}}G = G_{\text{m,B}} - G_{\text{m,A}} = \mu_{\text{B}} - \mu_{\text{A}} = \left(\frac{\partial G}{\partial \xi} \right)_{p,T}$$

- $\Delta_{\text{r}}G = 0$: reversible, equilibrium
- $\Delta_{\text{r}}G < 0$: irreversible, spontaneous reaction
- $\Delta_{\text{r}}G > 0$: impossible $\rightarrow$ spontaneous reverse reaction

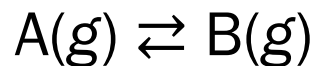
Progress of Reaction

$$G = \mu_A n_A + \mu_B n_B \quad \Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T}$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 6A.1)

Reaction Quotient



For a perfect gas, $\mu = \mu^\ominus + RT \ln(p/p^\ominus)$

$$\Delta_r G = \mu_B - \mu_A = \left\{ \mu_B^\ominus + RT \ln \left(\frac{p_B}{p_B^\ominus} \right) \right\} - \left\{ \mu_A^\ominus + RT \ln \left(\frac{p_A}{p_A^\ominus} \right) \right\}$$

Hence, using $\Delta_r G^\ominus = \mu_B^\ominus - \mu_A^\ominus$,

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\frac{p_B}{p_A} \right) = \Delta_r G^\ominus + RT \ln Q$$

Equilibrium Constant

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

Hence,

$$\Delta_r G^\ominus = -RT \ln K, \text{ where } K = (p_{\text{B}}/p_{\text{A}})_{\text{eq}}$$

Generalization

For an arbitrary reaction,

$$\sum_{\text{reactant } i} \nu_i \{i\} \rightleftharpoons \sum_{\text{product } j} \nu_j \{j\}$$



$$\nu_{\text{A}} = 2, \nu_{\text{B}} = 1, \nu_{\text{C}} = 3, \nu_{\text{D}} = 1$$

Now,

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

Generalization

$$dG = \sum_{\text{product } j} \mu_j dn_j + \sum_{\text{reactant } i} \mu_i dn_i$$

$$dG = \sum_{\text{product } j} \mu_j \nu_j d\xi - \sum_{\text{reactant } i} \mu_i \nu_i d\xi$$

$$dG = \left(\sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i \right) d\xi$$

$$\Delta_r G = \left(\frac{\partial G}{\partial \xi} \right)_{p,T} = \sum_{\text{product } j} \mu_j \nu_j - \sum_{\text{reactant } i} \mu_i \nu_i$$

Chemical Potential and Activity

- Real gas: $\mu = \mu^\ominus + RT \ln(f/p^\ominus)$
- Real solution (solvent): $\mu = \mu^* + RT \ln a$
- Real solution (solute): $\mu = \mu^\ominus + RT \ln a$
- Generally, let us use

$$\mu = \mu^\ominus + RT \ln a$$

Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

For each species x ,

$$\nu_x \mu_x = \nu_x (\mu_x^\ominus + RT \ln a_x) = \nu_x \mu_x^\ominus + RT \ln(a_x^{\nu_x})$$

Then

$$\begin{aligned} \sum_x \nu_x \mu_x &= \sum_x \nu_x \mu_x^\ominus + RT \sum_x \ln(a_x^{\nu_x}) \\ &= \sum_x \nu_x \mu_x^\ominus + RT \ln \prod_x a_x^{\nu_x} \end{aligned}$$

Generalization

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i$$

Substitution leads to

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \prod_{\text{product } j} a_j^{\nu_j} - RT \ln \prod_{\text{reactant } i} a_i^{\nu_i}$$

where $\Delta_r G^\ominus = \sum_{\text{product } j} \nu_j \mu_j^\ominus - \sum_{\text{reactant } i} \nu_i \mu_i^\ominus$

$$\Delta_r G = \Delta_r G^\ominus + RT \ln \left(\prod_{\text{product } j} a_j^{\nu_j} / \prod_{\text{reactant } i} a_i^{\nu_i} \right)$$

Q and K

$$\Delta_r G = \Delta_r G^\ominus + RT \ln Q$$

where the **reaction quotient** Q is defined as

$$Q = \frac{\prod_{\text{product } j} a_j^{v_j}}{\prod_{\text{reactant } i} a_i^{v_i}}$$

At equilibrium, $\Delta_r G = 0$

$$0 = \Delta_r G^\ominus + RT \ln Q_{\text{eq}} = \Delta_r G^\ominus + RT \ln K$$

where the **equilibrium constant** K is Q at equilibrium

What is $\Delta_r G^\ominus$?

$$\Delta_r G^\ominus = \sum_{\text{products } j} \nu_j \mu_j^\ominus - \sum_{\text{reactants } i} \nu_i \mu_i^\ominus$$

Here, $\mu_x^\ominus$ is the chemical potential of species x ...

- Gas: at standard pressure (1 bar)
- Solute/solvent: in its pure form

Thus, $\Delta_r G^\ominus$ is the **standard Gibbs energy of reaction**.

Connection to Thermochemistry

$$\Delta_r G^\ominus = -RT \ln K$$

Since we know

$$\Delta_r G^\ominus = \sum_{\text{products } j} \nu_j \Delta_f G^\ominus(j) - \sum_{\text{reactants } i} \nu_i \Delta_f G^\ominus(i),$$

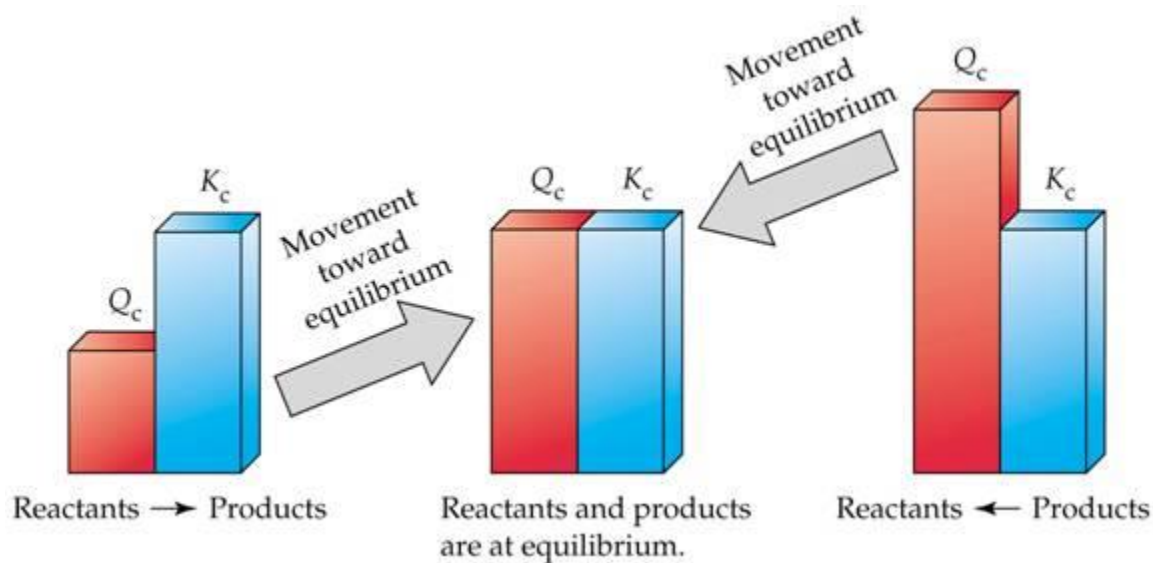
we can calculate K from thermochemical data.

Meaning of Q and K

- Reaction quotient Q
 - “How far the equilibrium is”
 - Varies as the reaction progresses
 - $Q = K$ at equilibrium
- Equilibrium constant K
 - Amounts of reactants and products at equilibrium
 - Constant for the given experimental condition

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Meaning of Q and K



Activity

$$\mu = \mu^\ominus + RT \ln a$$

- Gas: $\mu = \mu^\ominus + RT \ln(f/p^\ominus)$
- Solution
 - Solvent: $\mu_A = \mu_A^* + RT \ln a_A$
 - Solute: $\mu_B = \mu_B^* + RT \ln(K_B/p_B^*) + RT \ln a_B$
- Pure liquid and solid: ?

Approximation (perfect/ideal limit)?

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Approximate Activity of Gas

$$\mu = \mu^{\ominus} + RT \ln a = \mu^{\ominus} + RT \ln(f/p^{\ominus})$$

For a perfect gas,

$$\mu = \mu^{\ominus} + RT \ln(p/p^{\ominus})$$

Hence, approximately

$$a = \frac{p}{p^{\ominus}}$$

Approximate Activity of Solute

$$\mu = \mu^\ominus + RT \ln a$$

For an ideal-dilute solution,

$$\mu_B = \mu_B^* + RT \ln(K_B/p_B^*) + RT \ln x_B$$

$$x_B = \frac{n_B}{n_A + n_B} \approx \frac{n_B}{n_A} = \frac{V}{n_A} \frac{n_B}{V} = \frac{[B]}{[A]} = \frac{[B]}{c^\ominus} \frac{c^\ominus}{[A]}$$

Assuming that $[A]$ is constant,

$$\mu_B = \underbrace{\mu_B^* + RT \ln(K_B/p_B^*) - RT \ln([A]/c^\ominus)}_{\mu_B^\ominus} + RT \ln \underbrace{([B]/c^\ominus)}_a$$

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Approximate Activity of Solvent

$$\mu = \mu^\ominus + RT \ln a$$

For an ideal-dilute solution,

$$\mu_A = \mu_A^* + RT \ln x_A$$

$$x_A = \frac{n_A}{n_A + n_B} \approx \frac{n_A}{n_A} = 1$$

Hence, approximately

$$a = 1$$

Pure Solids and Liquids

$$\Delta G_{\text{m}} = \mu_f - \mu_i = \int_{p_i}^{p_f} V_{\text{m}} \, dp$$

Let the initial state to be the standard state:

$$\mu - \mu^{\ominus} = \int_{p^{\ominus}}^p V_{\text{m}} \, dp' \approx V_{\text{m}} \int_{p^{\ominus}}^p 1 \, dp'$$

From the definition of an activity, $\mu - \mu^{\ominus} = RT \ln a$,

$$\ln a = \frac{V_{\text{m}}}{RT} \int_{p^{\ominus}}^p 1 \, dp' = \frac{V_{\text{m}}}{RT} (p - p^{\ominus})$$

Example: Activity of Coke

The density of coke at 1000°C is about 1.5 g cm⁻³. Calculate the activity of C(s) in the form of coke at 100 bar and 1000°C.

The molar volume is

$$V_m = (\text{molar mass})/(\text{density}) = (12 \text{ g/mol})/(1.5 \text{ g cm}^{-3}) = 8.0 \text{ cm}^3 \text{ mol}^{-1}$$

Thus,

$$\begin{aligned}\ln a &= V_m(p - p^\ominus)/RT \\ &= (8.0 \text{ cm}^3 \text{ mol}^{-1}) \times (100 - 1) \text{ bar} / (0.082 \text{ L bar K}^{-1} \text{ mol}^{-1}) / (1273 \text{ K}) \\ &= 0.0076\end{aligned}$$

or $a = 1.01$.

Pure Solids and Liquids

$$\ln a = \frac{V_m}{RT} (p - p^\ominus)$$

At moderate pressures, the activity is very close to 1.

Hence, we usually take $a = 1$ for pure solids and liquids.

Approximation for Activity

State	Measure	Approximation for a_j	Definition
Solute	molality	$b_j/b^\ominus$	$b^\ominus = 1 \text{ mol kg}^{-1}$
	molar concentration	$[J]/c^\ominus$	$c^\ominus = 1 \text{ mol dm}^{-3}$
Gas phase	partial pressure	$p_j/p^\ominus$	$p^\ominus = 1 \text{ bar}$
Pure solid, liquid		1 (exact)	

For Gaseous Reactions

$$K_p = \frac{\prod_{\text{product } j} a_j^{\nu_j}}{\prod_{\text{reactant } i} a_i^{\nu_i}} = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/p^\ominus)^{\nu_i}}$$

Since $p_i/p^\ominus = [i](RT/p^\ominus) = [i]/c^\ominus (c^\ominus RT/p^\ominus)$,

$$K_p = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}} \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\sum_j \nu_j - \sum_i \nu_i}$$

$$K_p = K_c \left(\frac{c^\ominus RT}{p^\ominus} \right)^{\Delta \nu} = K_c \left(\frac{T}{12.03 \text{ K}} \right)^{\Delta \nu}$$

Equilibrium Constants

For gaseous reactions (assuming perfect gases),

$$K_p = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/p^\ominus)^{\nu_i}}$$

$$K_c = \frac{\prod_{\text{product } j} ([j]/c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i]/c^\ominus)^{\nu_i}}$$

Can you theoretically calculate their values?

Statistical Mechanics

$$\mu = \left(\frac{\partial A}{\partial n} \right)_{T,V} = -k_B T \left(\frac{\partial \ln Q}{\partial n} \right)_{T,V} = -RT \left(\frac{\partial \ln Q}{\partial N} \right)_{T,V}$$

For N perfect-gas molecules, $Q = \frac{1}{N!} q^N$ and

$$\ln Q \approx N \ln q - N \ln N + N$$

Hence,

$$\mu = -RT \ln q + RT \ln N + RT - RT = -RT \ln \left(\frac{q}{N} \right)$$

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Partition Functions in Action

$$\mu_x = -RT \ln \left(\frac{q_x}{N_x} \right)$$

At equilibrium,

$$\Delta_r G = \sum_{\text{product } j} \nu_j \mu_j - \sum_{\text{reactant } i} \nu_i \mu_i = 0$$

Substitution leads to

$$\sum_{\text{product } j} \nu_j \ln \left(\frac{q_j}{N_j} \right) = \sum_{\text{reactant } i} \nu_i \ln \left(\frac{q_i}{N_i} \right)$$

Tedious Math

$$\sum_{\text{product } j} v_j \ln \left(\frac{q_j}{N_j} \right) = \sum_{\text{reactant } i} v_i \ln \left(\frac{q_i}{N_i} \right)$$

$$\sum_{\text{product } j} \ln \left(\frac{q_j}{N_j} \right)^{v_j} = \sum_{\text{reactant } i} \ln \left(\frac{q_i}{N_i} \right)^{v_i}$$

$$\prod_{\text{product } j} \left(\frac{q_j}{N_j} \right)^{v_j} = \prod_{\text{reactant } i} \left(\frac{q_i}{N_i} \right)^{v_i}$$

$$\frac{\prod_{\text{product } j} N_j^{v_j}}{\prod_{\text{reactant } i} N_i^{v_i}} = \frac{\prod_{\text{product } j} q_j^{v_j}}{\prod_{\text{reactant } i} q_i^{v_i}}$$

Equilibrium Constant

$$\frac{\prod_{\text{product } j} N_j^{\nu_j}}{\prod_{\text{reactant } i} N_i^{\nu_i}} = \frac{\prod_{\text{product } j} q_j^{\nu_j}}{\prod_{\text{reactant } i} q_i^{\nu_i}}$$

$$\frac{\prod_{\text{product } j} \{N_j / (N_A V c^\ominus)\}^{\nu_j}}{\prod_{\text{reactant } i} \{N_i / (N_A V c^\ominus)\}^{\nu_i}} = \frac{\prod_{\text{product } j} \{q_j / (N_A V c^\ominus)\}^{\nu_j}}{\prod_{\text{reactant } i} \{q_i / (N_A V c^\ominus)\}^{\nu_i}}$$

The left-hand side is the equilibrium constant:

$$K_c = \frac{\prod_{\text{product } j} ([j] / c^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} ([i] / c^\ominus)^{\nu_i}} = \frac{\prod_{\text{product } j} \{q_j / (N_A V c^\ominus)\}^{\nu_j}}{\prod_{\text{reactant } i} \{q_i / (N_A V c^\ominus)\}^{\nu_i}}$$

Le Chatelier's Principle

When subjected to a disturbance,
a system at equilibrium tends to respond in a way
that **minimizes the effect of the disturbance**.

Disturbances to consider:

- Amount
- Volume & pressure (for a gaseous reaction)
- Temperature

Amount Change

- **Increase** of the amount of a particular species
 - The system responds to **decrease** the amount
- **Decrease** of the amount of a particular species
 - The system responds to **increase** the amount
- Reactant increase / product decrease: $Q < K$
 - Decrease of reactants & increase of products
- Reactant decrease / product increase: $Q > K$
 - Increase of reactants & decrease of products

Volume/Pressure Change

For a gaseous reaction,

- Increase of the volume → decrease of the pressure
- Decrease of the volume → increase of the pressure
- Pressure change due to addition of inert gas does not affect the equilibrium

$$K_p = \frac{\prod_{\text{product } j} (p_j/p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (p_i/p^\ominus)^{\nu_i}} = \frac{\prod_{\text{product } j} (n_j RT/V p^\ominus)^{\nu_j}}{\prod_{\text{reactant } i} (n_i RT/V p^\ominus)^{\nu_i}}$$

Volume/Pressure Change

$$K_p = \frac{\prod_{\text{product } j} (n_j RT / V p^\ominus)^{v_j}}{\prod_{\text{reactant } i} (n_i RT / V p^\ominus)^{v_i}}$$

For a volume change from V to xV ,

$$Q_p = \frac{\prod_{\text{product } j} (n_j RT / xV p^\ominus)^{v_j}}{\prod_{\text{reactant } i} (n_i RT / xV p^\ominus)^{v_i}}$$
$$Q_p = K_p x^{-(\sum_j v_j - \sum_i v_i)} = K_p x^{-\Delta v}$$

Volume/Pressure Change

$$Q_p = K_p x^{-\Delta\nu}$$

For $x > 1$ (volume increase),

- $\Delta\nu = 0$: $Q_p = K_p$ and the system is at equilibrium
- $\Delta\nu > 0$: $Q_p < K_p$ and the forward reaction is favored
- $\Delta\nu < 0$: $Q_p > K_p$ and the reverse reaction is favored

Temperature Change

- Exothermic reactions
 - Increased temperature favors the reactants
 - Decreased temperature favors the products
- Endothermic reactions
 - Increased temperature favors the products
 - Decreased temperature favors the reactants

van't Hoff Equation

$$\ln K = -\Delta_r G^\ominus / RT$$

Differentiating both sides with respect to T ,

$$\frac{d \ln K}{dT} = -\frac{1}{R} \frac{d(\Delta_r G^\ominus / T)}{dT}$$

Using the Gibbs-Helmholtz equation,

$$-R \frac{d \ln K}{dT} = \frac{d(\Delta_r G^\ominus / T)}{dT} = -\frac{\Delta_r H^\ominus}{T^2}$$

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

T*-Dependence of *K

$$\frac{d \ln K}{dT} = \frac{\Delta_r H^\ominus}{RT^2}$$

Integration gives

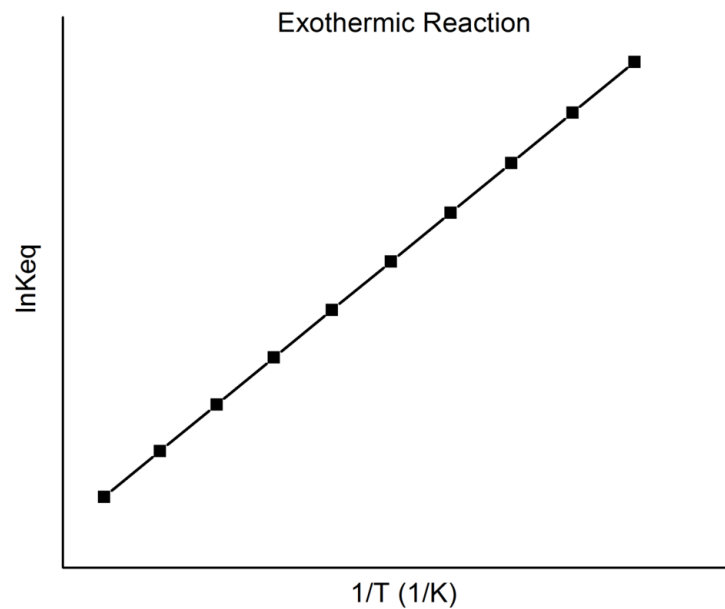
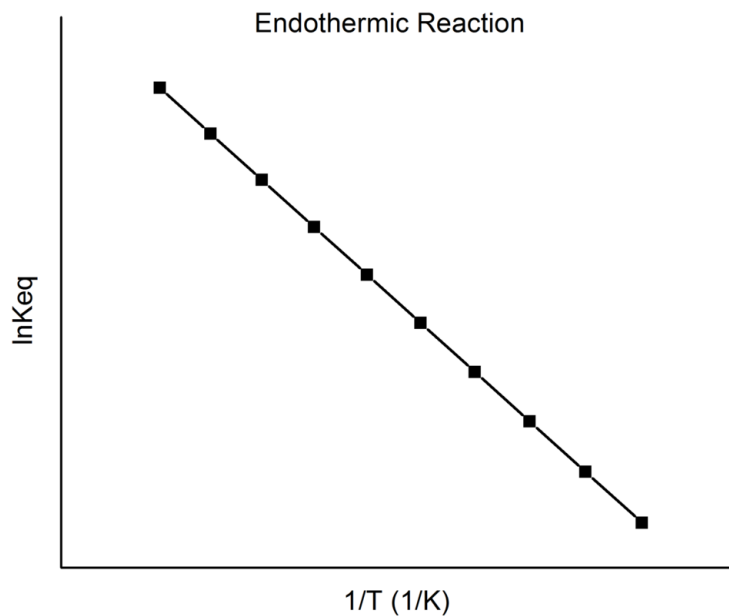
$$\int_{\ln K_1}^{\ln K_2} d \ln K = \int_{T_1}^{T_2} \frac{\Delta_r H^\ominus}{RT^2} dT$$

Assuming that $\Delta_r H^\ominus$ is constant,

$$\ln K_2 - \ln K_1 = \frac{\Delta_r H^\ominus}{R} \int_{T_1}^{T_2} \frac{1}{T^2} dT = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$

T-Dependence of *K*

$$\ln K_2 - \ln K_1 = -\frac{\Delta_r H^\ominus}{R} \left(\frac{1}{T_2} - \frac{1}{T_1} \right)$$



Summary

